

① Train Accuracy $\rightarrow 78.48\%$ (95% accuracy)

② Test Accuracy $\rightarrow 70.40\%$ (65% accuracy)

Overfitting $\rightarrow$ Model generalises the output because it is over-trained on training set and hence performed well on training set but unable to perform on test set.

Underfitting $\rightarrow$ Model unable to understand hidden patterns and shows less accuracy while training and testing.

Overfitting $\rightarrow$ Overfitting occurs when a model is excessively complex - having too many parameters or learning for too long. Instead of learning general rules, it memorizes everything on training data including random noise and outliers.

[High Variance]

$\rightarrow$ Symptoms $\rightarrow$ Exceptionally low error [high accuracy] on the training set, but high error [low accuracy] on testing/validation set.

$\rightarrow$ Common Cause $\rightarrow$ Training on a small data or high noise data

Training on too many epochs without a stopping mechanism.

→ Fix Overfitting →

① Regularization Add penalty to the model parameters [L1/lasso or L2/ridge] to prevent weights from becoming too large

② Simplify the Model

③ Cross Validation → k fold

④ Early Stopping →

② Underfitting → Underfitting happens when a model is too simple to capture the underlying structure of the data. It makes strong, incorrect assumptions [eg trying to fit straight line to a highly curved, complex dataset]

[High Bias]

Common Causes →

- ① Using an algorithm that is too simple for complex non-linear data

- ② Insufficient training time

- ③ Having very few input features

Fix Underfitting →

① Increase the model complexity: Switch to a complex algorithm and add more layers to neural network or use polynomial features

② Train longer